

Lagrangian Mean Curvature Flow

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Abstract

The main goal of these lecture notes is to introduce Lagrangian mean curvature flow. The first couple of sections explain the basic definitions and key examples, such as the Lagrangian catenoid, Clifford tori, and Whitney spheres. In the third section, we study singularity models, tangent flows, and Type II blow-downs. We conclude by discussing open problems in the field.

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1 Lecture I: Introduction

Definition 1.1 (Mean curvature flow). Let L_t^n be an immersed or embedded 1-parameter family of n -dimensional submanifolds in $\mathbb{R}^N$, then we say L_t^n flows by mean curvature if the following equation holds

$$\frac{\partial L_t}{\partial t} = H_t \quad (1)$$

where H_t is the mean curvature of L_t^n .

We will write MCF for mean curvature flow. MCF is the negative gradient flow of the volume functional, and hence gives the steepest descent of volume. A major difficulty is that MCF is much less well behaved in higher codimension, namely when the ambient dimension is greater than $n + 1$. One way to deal with this issue is to study the flow within special classes of submanifolds that are preserved under MCF.

1.1 Lagrangian mean curavture flow

Let $N = 2n$. Then $\mathbb{R}^{2n}$ can be identified with $\mathbb{C}^n$. We write a point in $\mathbb{C}^n$ as $(z_1, \dots, z_n)$, where $z_k = x_k + iy_k$ for $1 \leq k \leq n$. We begin by recalling some basic definitions.

Definition 1.2 (Kähler form). The Kähler form ω on $\mathbb{C}^n$ is defined by

$$\omega := dx_1 \wedge dy_1 + \dots + dx_n \wedge dy_n \quad (2)$$

$$= \frac{i}{2} (dz_1 \wedge d\bar{z}_1 + \dots + dz_n \wedge d\bar{z}_n) \quad (3)$$

Definition 1.3 (Complex structure). In $\mathbb{R}^{2n}$, a linear almost complex structure is an endomorphism J such that $J^2 = -\text{Id}$. The linear group $GL(2n, \mathbb{R})$ acts on the space of these structures by conjugation: $g \cdot J = gJg^{-1}$.

Definition 1.4 (Lagrangian submanifolds). Let L be an n -dimensional submanifold of $\mathbb{C}^n$, then L^n is said to be Lagrangian iff $\omega|_L = 0$.

Remark 1.5. If $\tilde{L}$ is a k -dimensional subspace of $\mathbb{C}^n$ such that $\omega|_{\tilde{L}} = 0$, then $k \leq n$. Thus, a Lagrangian submanifold is a maximal isotropic submanifold, meaning that its dimension is as large as possible among submanifolds on which ω vanishes.

Exercise 1.6. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a function *regularity?* and let $\Gamma_F = \{x + iF(x) \mid x \in \mathbb{R}^n\}$. Then show that Γ_F is Lagrangian iff there exists a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, such that $F = \nabla f$.

Definition 1.7 (Volume form). On $\mathbb{C}^n$, we define the volume form Ω as follows

$$\Omega = dz_1 \wedge \dots \wedge dz_n \quad (4)$$

Lemma 1.8. Let $L^n \hookrightarrow \mathbb{C}^n$ be an oriented submanifold, then $|\Omega|_{L^n} \leq 1$ and the equality holds iff L^n is Lagrangian.

Proof. Let P be a n -plane in $\mathbb{C}^n$, let $\{e_1, \dots, e_n\}$ be an ONB of P . Then there exists an $A \in GL(n, \mathbb{C})$ such that if $\{u_1, \dots, u_n\}$ be the standard basis of $\mathbb{R}^n \hookrightarrow \mathbb{C}^n$, $e_j = \Delta u_j$ for all j . Now

$$\begin{aligned} |\Omega(e_1, \dots, e_n)|^2 &= |\Omega(Au_1, \dots, Au_n)|^2 \\ &= |\det_{\mathbb{C}}(A)|^2 \\ &= |\det_{\mathbb{R}}(A)|^2 \\ &= |e_1 \wedge J e_1 \wedge \dots \wedge e_n \wedge J e_n| \\ &\leq |e_1| |J e_1| \dots |e_n| |J e_n| \\ &= 1 \end{aligned}$$

Thus we proved the first part, for the second part, observe that the equality holds iff $e_j \perp J e_j$ for all j which implies P is Lagrangian. $\square$

Definition 1.9 (Lagrangian angle). Let $L^n \hookrightarrow \mathbb{C}^n$ be an oriented Lagrangian submanifold. The restriction of the volume form to L can be written as $\Omega|_L = e^{i\theta} \text{vol}|_L$, where $e^{i\theta} : L \rightarrow \mathbb{S}^1$, is called the Phase map. We call θ the Lagrangian angle.

Remark 1.10. Observe that the Lagrangian angle θ need not be single-valued, can be multivalued. For example, consider the unit circle $S^1 \subset \mathbb{C}$, parametrized by

$$z(\varphi) = e^{i\varphi}.$$

Since $\Omega = dz$, we have

$$dz|_{S^1} = ie^{i\varphi} d\varphi = e^{i(\varphi+\pi/2)} d\varphi.$$

Thus, the phase map is

$$e^{i\theta} = e^{i(\varphi+\pi/2)}.$$

Thus, we can choose

$$\theta = \varphi + \frac{\pi}{2}.$$

After going once around the circle, θ changes by 2π , this in general, θ is defined only up to multiples of 2π , but the phase function is well defined.

The Lagrangian angle and mean curvature are closely related. Indeed, we have the following proposition

Proposition 1.11. Let $L \subset \mathbb{C}^n$ be a Lagrangian submanifold with Lagrangian angle θ . Then

$$H = J\nabla\theta. \tag{5}$$

An easy consequence of the above proposition is the following characterisation of minimal ($H = 0$) Lagrangians.

Proposition 1.12. Let $L \subset \mathbb{C}^n$ be a connected Lagrangian submanifold. If L is minimal, then θ is a constant.

Definition 1.13 (Special Lagrangian). Let $L^n \hookrightarrow \mathbb{C}^n$, is called special Lagrangian (with phase $e^{i\theta_0}$) if $\theta = \theta_0$ for some $\theta_0 \in \mathbb{R}$.

Proposition 1.14. The following are equivalent

1. L is special Lagrangian with phase $e^{i\theta_0}$
2. $\omega|_L = 0$ and $\text{Im}(e^{-i\theta_0}\Omega|_L) = 0$
3. $\text{Re}(e^{-i\theta_0}\Omega|_L) = \text{Vol}(L)$

Remark 1.15. Connected special Lagrangians are minimal ??

Remark 1.16. Everything so far works if we replace $\mathbb{C}^n$ by (M, J) , where M is n -dimensional complex manifold and J complex structure with ω and Ω such that for all $p \in M$, there exists an isomorphism (of complex vector space)

$$i_p : T_p M \rightarrow \mathbb{C}^n \quad (6)$$

such that

$$\begin{aligned} i_p^* \omega_{\mathbb{C}^n} &= \omega(p) \\ i_p^* \Omega_{\mathbb{C}^n} &= \Omega(p) \end{aligned}$$

such a (M, J, ω, Ω) defines a metric as follows

$$g(JX, Y) = \omega(X, Y) \quad (7)$$

and M is called Calabi-Yau. This also implies g is Kähler and $\text{Ric} = 0$.

Proposition 1.17. Let L be a compact special Lagrangian in M , suppose also that M is Calabi-Yau. Then L minimises volume in $[L] \in H_n(M)$.

Proof. Suppose $\theta_0 = 0$, i.e $\text{Re}(\Omega|_L) = \text{vol}|_L$. Let $L^1 \in [L]$, then

$$\text{vol}(L) = \int_L \text{vol}|_L = \int_L \text{Re}(\Omega) = [L][\text{Re}\Omega] \quad (8)$$

□

Now we ask a natural question here, which classes in $H_n(M)$ admit a special Lagrangian representative.

Recall that, in the introduction, we discussed that MCF behaves poorly in higher codimension, and one suggestion to overcome this problem was to study a special class of solutions. Now that we have defined the Lagrangian submanifolds of Calabi-Yau manifolds, it is natural to ask the following question: if we start with a Lagrangian submanifold L_0 , can we deform it through a family L_t such that each L_t remains Lagrangian? In fact, this is possible. If you start with a compact Lagrangian in Calabi-Yau manifold, then MCF exists for a short time, and the Lagrangian condition is preserved under the flow. We have the following theorem by Smoczyk.

Theorem 1.18 (Smoczyk, [Smo96]). *Let L_0 be a compact Lagrangian submanifold of the Calabi–Yau manifold M . Then there exists a unique, smooth solution $(L_t)_{t \in [0, T]}$ to the MCF starting with L_0 . Moreover, as long as the solution exists, all submanifolds L_t are Lagrangian submanifolds of M .*

Proof. To see that the Lagrangian condition is preserved under the MCF, denote $f_t := |\omega_t|^2$, then there exists a $c > 0$ such that

$$\frac{\partial f_t}{\partial t} \leq \Delta f_t + c f_t. \quad (9)$$

Now the result follows from applying the parabolic maximum principle. $\square$

The above theorem motivates the definition of Lagrangian mean curvature flow.

Definition 1.19 (Maslov class). *Let L be a Lagrangian with Lagrangian angle θ , then we define the Maslov class to be the cohomology class $[d\theta] \in H^1(L)$. If $d\theta = 0$, then we call it zero Maslov.*

Proposition 1.20. *LMCF preserves the zero Maslov condition.*

Lemma 1.21. *Under zero Maslov LMCF we have*

$$\frac{\partial \theta_0}{\partial t} = \Delta_t \theta_t \quad (10)$$

Corollary 1.22. *The only smooth zero Maslov self-shrinker in C^n is a plane. in any codimension?*

Exercise 1.23. *Let $f_t : \mathbb{R}^n \rightarrow \mathbb{R}$ then we consider $L_t = \{x + i \nabla f_t(x) : x \in \mathbb{R}^n\}$. Show that,*

$$\frac{\partial L_t}{\partial t} = H_t \iff \frac{\partial f_t}{\partial t} = \sum_{j=1}^n \arctan(\lambda_j(t))$$

where λ_j 's are the eigenvalues of the Hessian of f_t .

2 Lecture II: Examples

Picturing Lagrangian submanifolds is difficult since they always live in a space twice the dimension. However, we can consider some examples. The simplest examples are the curves on surfaces.

2.1 The Lagrangian catenoid

In [CU99] I. Castro and F. Urbano introduced the Lagrangian catenoid on $\mathbb{C}^n$.

Let $f(x_1, x_2) = \frac{1}{2} \log(x_1^2 + x_2^2)$ where $(x_1, x_2) \in \mathbb{R}^2 \setminus \{0\}$. Then we consider L be the graph of the gradient of f , $L = \Gamma_{\nabla f}$,

$$L := \left\{ \left(x_1 + i \frac{x_1}{x_1^2 + x_2^2}, x_2 + i \frac{x_2}{x_1^2 + x_2^2} \right) : (x_1, x_2) \neq 0 \right\}.$$

Since, f a harmonic function, $\Delta f = 0$, then it special Lagrangian with $\theta = 0$.

The behaviour of L on the extremal case shows the following,

- If $|(x_1, x_2)| \rightarrow \infty$ then L looks like $\{(x_1, x_2) \in \mathbb{R}^2 : (x_1, x_2) \neq 0\}$.
- If $|(x_1, x_2)| \rightarrow 0$ then L looks like $L \approx i\mathbb{R}$.

In a more general case, given $(\theta_1, \dots, \theta_n) \in (0, \pi)^n$ such that $\sum_{i=1}^n \theta_i = \pi$. Then, we consider the following,

- $P_1 = \mathbb{R}^n$.
- $P_2 = \{(e^{i\theta_1}x_1, \dots, e^{i\theta_n}x_n) : (x_1, \dots, x_n) \in \mathbb{R}^n\}$.

There, there exists an special Lagrangian with asymptotic planes $P_1 \cup P_2$ and Lagrangian angle $\theta = 0$ named the Lawlor neck.

2.2 The Clifford tori

Let $L = \{\sqrt{2}(e^{i\theta_1}, e^{i\theta_2}) : \theta_1, \theta_2 \in \mathbb{C}\} \subset \mathbb{S}^3(2)$. Then L is a Lagrangian manifold.

Exercise 2.1. *Compute the Lagrangian angle of the Clifford tori.*

The answer is $\theta = \theta_1 + \theta_2 + \pi$. Then L is not 0-Maslov since $[d\theta] \neq 0 \in H^1(L) \cong \mathbb{Z}^2$. Moreover, the mean curvature of the Clifford tori satisfies that,

$$H = \frac{x^\perp}{2},$$

where x is the position vector. Thus, L is a self-shrinker of the Lagrangian Mean Curvature Flow.

2.3 Lagrangian spheres

A fact due to Gromov [Gro85] is that does not exists embedded Lagrangian spheres in $\mathbb{C}^n$. However, there exists immersed Lagrangian spheres. Let

$$L := \left\{ \frac{(u_1 + iu_1u_3, u_3, u_2 + iu_2u_3)}{1 + u_3^2} : u_1^2 + u_2^2 + u_3^2 = 1 \right\}$$

be an immersion of the unit sphere $\mathbb{S}^2$ unit sphere centered at the origin of $\mathbb{R}^3$ into $\mathbb{C}^2$.

Exercise 2.2. *Show that L is immersed with a single transversal double point at 0.*

Comment 2.3. *L is zero Maslov.*

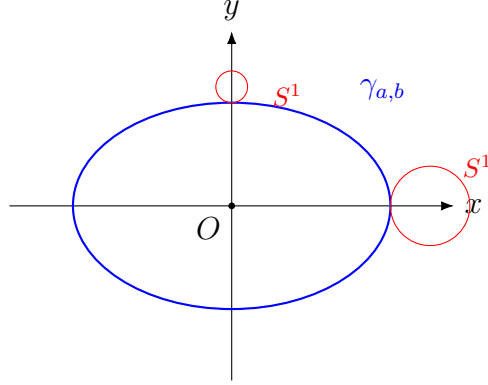


Figure 1

2.4 Equivariant Lagrangian Mean Curvature Flow in $\mathbb{C}^2$

For simple higher-dimensional examples, we consider equivariant submanifolds. These are Lagrangian submanifolds that can be described as the orbit of a profile curve $\gamma \subset \mathbb{C}$. More precisely, we define the associated circle invariant Lagrangian surface by

$$L_\gamma = \{(\gamma(s) \cos \phi, \gamma(s) \sin \phi) : \phi \in [0, 2\pi]\} \subseteq \mathbb{C}^2.$$

Under mean curvature flow, instead of studying the evolution of the entire Lagrangian surface, one may study only the evolution of the profile curve.

Therefore, the previous examples, Lagrangian tori, spheres and catenoids can be seen as equivariant Lagrangians, where the profile curve is a embedded closed curve, immersed closed curve, or a embedded open curve respectively.

Exercise 2.4. Let L_γ be an equivariant Lagrangian surface in $\mathbb{C}^2$. Show that the mean curvature flow of L_γ is given by

$$\frac{\partial L_\gamma}{\partial t} = H \leftrightarrow \frac{\partial \gamma}{\partial t} = \kappa - \frac{\gamma^\perp}{|\gamma|^2}$$

1. Let $\gamma_{a,b}$ be an ellipse with semi-axes a and b , $a \geq b > 0$, centered at the origin. Then L_γ is the **Lagrangian torus** for $\gamma_{a,b}$ for every point in the curve we have a circle in the Lagrangian, where the size of the circle is equal to the distance from the origin at that point of the curve. If we take $a = b$ in the figure then we get a circle and the corresponding Lagrangian is known as the torus. In this case, the circles at each point are all the same size. If $a = b = \sqrt{2}$ then we have the Clifford torus.
2. The **Lawlor neck** is an equivariant special Lagrangian with Lagrangian angle $\bar{\theta}$. Its profile curve may be expressed in polar coordinates as

$$r(\alpha) = \frac{B}{\sqrt[n]{\sin(\bar{\theta} - n\alpha)}}$$

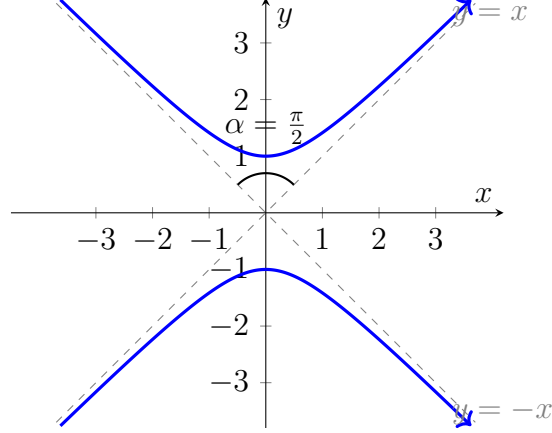


Figure 2: A hyperbola curve defining the Lawlor neck

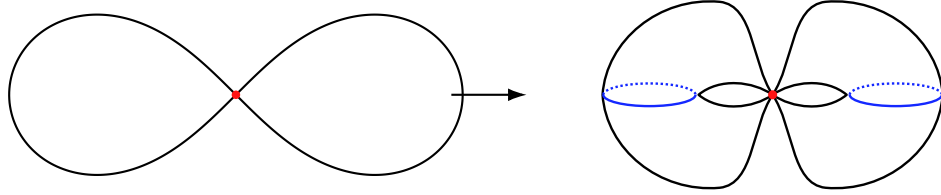


Figure 3: The figure eigh, on the left side, defines the Whitney sphere on $\mathbb{C}^2$.

for $B \geq 0$. Given any pair of equivariant planes that span an angle of $\frac{\pi}{n}$, there exists a Lawlor neck (unique up to scaling) asymptotic to these planes. For $n = 2$ and $\bar{\theta} = \frac{\pi}{2}$, the profile curve is the hyperbola, in the figure 2, which may be parametrised by

$$\sigma : \mathbb{R} \rightarrow \mathbb{C}, \quad \sigma(s) := (\sinh(s) \cosh(s)).$$

3. If we have that γ is a figure eight which passes through the origin, as in the figure 3, then we define the the Lagrangian surface L_γ to be the **Witney sphere**. In this case, the circles at each point of the curve shrink to a point as we approach the origin.

2.5 Evolution under the Lagrangian Mean Curvature flow

What does the Lagrangian Mean Curvature Flow do to L_γ ? Under mean curvature flow, we may study the evolution of the profile curve γ instead of the entire Lagrangian, which means that equivariant Lagrangian mean curvature flow can be studied in a similar way to curve shortening flow. The study of the evolution of the equivariant Lagrangian mean curvature flow is carried out in [Nev13; Nev10; GSS+07].

2.5.1 Torus case

- If γ is a circle, then L_γ is a torus. Under LMCF, the torus shrinks to a point, as it can be seen in the figure 4. Therefore, the Clifford torus has a Type I singularity.

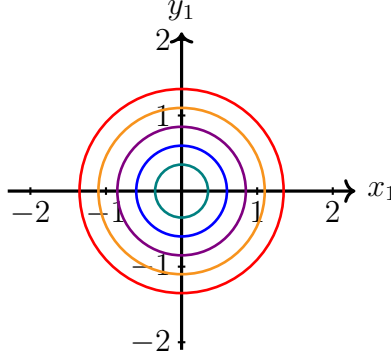


Figure 4

- If $a \neq b$. In [ELS20] the Lagrangian Mean Curvature Flow starting from the Lagrangian torus $L_{\gamma_{a,b}}$ has a finite time singularity which is not modeled on the Clifford torus. Then $a \gg b$ the torus will develop a Type II singularity and the curve at the singular time will look like a figure eight, as shown in [Nev10; GSS+07]. Thus, the Lagrangian torus will become a 2-sphere at the singular time. However, if a is not too big compared to b and $a \cdot b = 1$, then the Lagrangian Mean Curvature Flow starting from this initial data cannot shrink to a point [ELS20].

The Chekanov torus is an example of a Lagrangian torus whose profile curve is not a circle enclosing the origin, see figure ?? . It is known that the Chekanov torus generated by a curve not enclosing the origin, see figure 5. In [Gro07] K. Groh shows that the Lagrangian Mean Curvature Flow starting from the Chekanov torus contract to a point outside the origin.

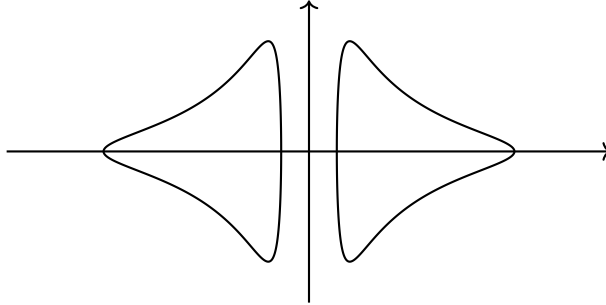


Figure 5: A curve defining a Chekanov torus T^2 in $\mathbb{C}^2$

2.5.2 Spheres

- If the profile curve is the figure eight can not be self-similar as there are no lagrangian self-shrinkers sheres. However, it will still shrink to a point in finite time, as shown in [SS19; Via21; Woo24]. At the finite time singularity the Lagrangian mean curvature flow will have a Type II singularity whose:
 - Type I blow-up is a plane with multiplicity two.

- Type II blow-up is the product of a Grim Reaper curve with a line.

Along the flow it deforms as in Figure 6, “squashing” vertically faster than it does horizontally, and one can imagine the Grim Reaper curve emerging at the extreme left (and right) points of the curve in the limit.

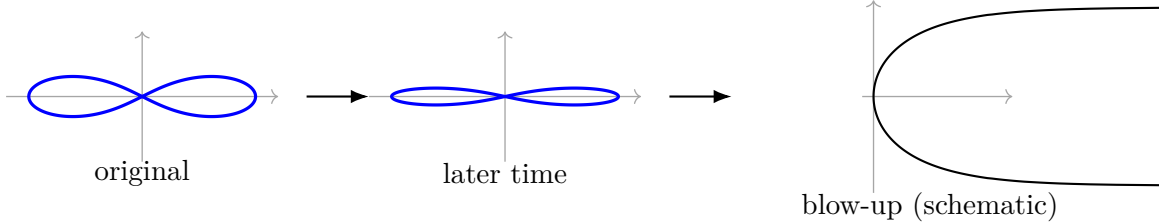


Figure 6: A collapsing figure eight along the Lagrangian mean curvature flow

- If the curve γ is contained in the region defined by the dashed lines, which each make an angle of $\frac{\pi}{4}$ with the horizontal axis, and γ meets any circle centred at the origin in at most 4 points. Then, in [Via21] Viana shows that the Lagrangian mean curvature flow starting from γ has a finite-time singularity at the origin where the flow shrinks to a point. The corresponding Lagrangian mean curvature flow has a Type II singularity at the origin, whose Type I blow-up is a plane with multiplicity two (roughly, two copies of the same plane).
- If the Ricci curvature of the induced metric on L satisfies $\text{Ric} \geq cr^2$ where $c > 0$ is a constant and r is the distance to the origin in $\mathbb{C}^2$. Then, in [SS19] Savas-Halilaj shows that the Lagrangian mean curvature flow starting from L has a finite-time singularity at the origin where the flow shrinks to a point. Moreover, it has a Type II singularity at the origin whose Type II blow-up is the product of a Grim Reaper curve and a real line.

2.5.3 Cylinders

In the case that the profile curve is an open curve, then the behaviour of the Lagrangian mean curvature flow will be determined by the angle of the curves α , see figure 2.

In the case where $\alpha = \pi/2$ then the equivariant Lagrangian surface will be the Lawlor neck or Lagrangian catenoid. Then, the corresponding Lagrangian will be minimal, $H = 0$. Thus, it will be a stationary solution of the Lagrangian mean curvature flow.

Remark 2.5. In [Su19] Su considered a family of curves γ asymptotic to a pair of lines with angle $\alpha = \frac{\pi}{2}$ that are trapped between hyperbolae defining Lawlor necks. Su showed that the Lagrangian mean curvature flow with initial data one of the elements of this family exist for all time and converges to a Lawlor neck.

If $\alpha \in (0, \pi/2)$ then in [Anc] showed that there is a dilation family of curves γ so that the corresponding Lagrangian is a *self-expander*.

Remark 2.6. In [Su19] Su considered certain natural curves γ for $\alpha \in (0, \frac{\pi}{2})$, bounded between two curves defining self-expanders, and showed that (after rescaling) Lagrangian mean curvature flow starting at the Lagrangian defined by γ exists for all time and converges to an Anciaux self-expander.

If $\alpha \in (\frac{\pi}{2}, \pi)$, it is shown in [Nev; Woo24] that for curves α the flow has a first finite-time singularity at the origin. The singularity is Type II and the Type I blow-up is a circle-invariant pair of Lagrangian planes defined by a pair of lines in $\mathbb{C}$ which meet at right angles at 0. Wood [Woo24] has shown further that the Type I blow-up at the origin is unique, the Type II blow-up is unique (up to scale) and given by the Lawlor neck asymptotic to the Type I blow-up, so that the blow-down of the Type II blow-up is equal to the Type I blow-up.

3 Lecture III: Singularities

In this section, we discuss the singularities of LMCF. It is well known that, in MCF, ancient solutions arise as type-II blow-ups of singularities, as proved in [HS99]. An early result of Neves [Nev07] shows that, in the zero-Maslov-class case, the tangent flow at a singularity at the origin. Later, Lambert, Lotay, and Schulze [LLS21] studied blow-downs of ancient solutions, with the aim of characterizing the singularity models.

3.1 Definitions and Examples

First, we recall the definition of the maximal time of existence, which is the largest time up to which the flow exists smoothly. If $T < \infty$, then the flow develops a singularity at time T . Moreover, the curvature blows up near the singular time T .

Theorem 3.1. *Let $T < \infty$ be the maximal time of existence for a LMCF (L_t) , then $\sup_{L_t} |A_t|^2 \rightarrow \infty$ as $t \rightarrow T$, where A_t is the second fundamental form of L_t .*

One can classify the singularities according to the rate at which the curvature blows up.

Definition 3.2 (Type I and Type II singularities). *Let $T < \infty$ be the first singular time of a Lagrangian mean curvature flow (L_t) . We say that the flow develops a Type I singularity as $t \nearrow T$ if there exists a constant $C > 0$ such that*

$$\max_{L_t} |A_t|^2 \leq \frac{C}{T - t}. \quad (11)$$

A singularity is of type II if it is not of type I.

One way to understand the singularity is to zoom in near the singular point and study the limiting flow. More precisely, one can rescale the flow parabolically around a point (X_0, T) and any limit obtained in this way is called a tangent flow.

Definition 3.3 (Tangent flow). *Let $(L_t)_{(0 < t < T)}$ be a LMCF, with maximal time of existence T . Let $x_0 \in \mathbb{C}^n$, and $\sigma_j > 0$ be a sequence such that $\sigma_j \rightarrow \infty$, for $s < 0$, define the parabolically rescaled flows by*

$$L_s^j = \sigma_j \left(L_{T + \sigma_j^{-2}s} - x_0 \right) \quad (12)$$

then after passing to a subsequence, the flow (L_s^j) converges to a limiting flow L_s^∞ , which is known as the tangent flow of (L_t) at (x_0, T) .

The tangent flow, obtained by the parabolic rescaling at a fixed singular time T , is always an ancient self-shrinking solution, for example, see [CM12]. Note that the tangent flow depends on the sequence we choose, and thus may not be unique.

To understand the higher-order information of the singularity, we consider the type-II blow-up, where we consider rescaling around the points of high curvature.

Definition 3.4 (Type II blow up). Let $(L_t)_{(0 < t < T)}$ be an LMCF, with maximal time of existence T . Let $t_j \nearrow T$ and $x_j \in L_{t_j}$, define σ_j as follows

$$\sigma_j := |A(x_j, t_j)| = \sup\{A(x, t) | t \leq t_j, x \in L_t\}. \quad (13)$$

Then consider the rescaled flow

$$L_s^j = \sigma_j \left(L_{t_j + \sigma_j^{-2}s} - x_j \right). \quad (14)$$

If a subsequence (L_s^j) converges to a limiting flow (L_s^∞) , then the limit is called the type II blow-up of the flow.

The type II blow-ups are ancient solutions of the flow. One can also consider the blow-down of the flow.

Definition 3.5 (Blow-down). Let $(L_t)_{-\infty < t < 0}$ be an ancient solution to LMCF, let $\sigma_j > 0$ such that $\sigma_j \rightarrow 0$. Consider the rescaled flow

$$L_s^j = \sigma_j^{-1} L_{\sigma_j^2 s}. \quad (15)$$

If a subsequence (L_s^j) converges to a limiting flow (L_s^∞) , then limit is called the blow-down of the flow.

Using Huisken's monotonicity formula [Hui90], one can say that the limiting surface should be an ancient solution, especially a self shrinker.

Example 3.6. Consider the Clifford torus in $\mathbb{C}^n$, under LMCF, the Clifford torus shrinks homothetically to the origin in finite time. Thus the tangent flow of the Clifford torus at the origin is the self-shrinker Clifford torus itself, while the tangent flow of any other point is empty.

Example 3.7. Recall, the Whitney sphere also collapses to origin in finite time, but not a self-shrinker. The tangent flow at the origin is Lagrangian plane with multiplicity 2, see [Via21]. The singularity is type II, and the Type II blow-up is the product of Grim reaper curve with flat Lagrangian plane, see [SS19], with blow-down Lagrangian plane with multiplicity 2.

Example 3.8. Lagrangian cone

One interesting observation is that, in the examples discussed above, the tangent flow at the finite-time singularity and the blow-down of the corresponding ancient Type II blow-up appear to have the same asymptotic structure. We expect this phenomenon to hold more generally, although it has not yet been proved. Nevertheless, we record two separate results concerning the structure of tangent flows and blow-downs in the zero-Maslov-class setting.

3.2 Results about the classification of singularities

The first result is about the structure of tangent flows of zero Maslov LMCF.

Theorem 3.9 (Neves, [Nev07]). *Let (L_t) be a zero Maslov LMCF with finite time singularity. Then any tangent flow is a finite union of special Lagrangian C_j with Lagrangian angle θ_j , and θ_j 's do not depend on the choice of the tangent flow.*

Note that the Clifford torus does not contradict the above theorem, because it does not have a zero Maslov class.

Theorem 3.10 (Lambert-Lotay-Schulze, [LLS21]). *Let $(L_t)_{-\infty < t < 0}$ be an ancient solution to LMCF, for any blow-down sequence, after passing to the subsequence, the blow-down limit is union of special Lagrangian cones, and the Lagrangian angles of these cones are independent of the blow-down sequence.*

Observe that the two results above do not imply that the tangent flow and the blow-down of a corresponding ancient Type II blow-up are the same. Indeed, even if both limits are unions of Lagrangian cones, their Lagrangian angles need not agree.

We now focus on the two-dimensional case, where one can say more about the structure of tangent flows.

Lemma 3.11. *Special Lagrangian cone in $\mathbb{C}^2$ is union of planes.*

Proof. □

Corollary 3.12. *For a zero-Maslov-class LMCF of surfaces in $\mathbb{C}^2$, any tangent flow is a finite union of planes P_j , where each P_j is a special Lagrangian plane, possibly with different phases.*

The simplest case is when the tangent flow consists of two planes, say P_1 and P_2 . Depending on how P_1 and P_2 intersect, there are three possible cases. Before studying these cases, we introduce some important quantities that are invariant under LMCF.

Definition 3.13 (Almost calibrated). *The Lagrangian manifold L is called almost calibrated if the following condition holds for the Lagrangian angle*

$$\sup_L \theta - \inf_L \theta < \pi \tag{16}$$

One motivation behind defining the almost calibrated Lagrangian is to avoid surfaces like grim reaper cylinder. Note that the grim reaper is not almost calibrated, but it is an example where it's sharp. But note that grim reaper cylinder is zero Maslov.

Proposition 3.14. *Almost calibrated condition is preserved under the flow*

Proof. This follows from the fact that the Lagrangian angle satisfies the Heat equation

$$\frac{\partial \theta}{\partial t} = \Delta \theta \tag{17}$$

□

Let $\mathbb{C}^n$ be endowed with standard complex coordinates $(z_1 = x_1 + iy_1, \dots, z_n = x_n + iy_n)$, recall the Liouville form λ as follows

$$\lambda = \sum_{j=1}^n x_j dy_j - y_j dx_j. \quad (18)$$

Note that $d\lambda = 2\omega$. Thus if L is Lagrangian, then by definition $\omega|_L = 0$, hence

$$d(\lambda|_L) = (d\lambda)|_L = 2\omega|_L = 0.$$

Thus $\lambda|_L$ is a closed one-form on L , and so it determines a cohomology class $[\lambda|_L] \in H^1(L; \mathbb{R})$. This motivates the following definition.

Definition 3.15. *A Lagrangian $L \in \mathbb{C}$ is called exact if the Liouville form λ restricted to L is exact, i.e., $[\lambda|_L] = 0$.*

Exercise 3.16. *Suppose L is given as follows*

$$\{x + i\nabla f(x) \mid x \in \mathbb{R}\}, \quad (19)$$

then show that L is exact.

We now study the three possible cases in the two-dimensional setting.

Case 1 $P_1 \cap P_2 = \{0\}$. In this case we have the following theorem

Theorem 3.17 (Lambert-Lotay-Schulze, [LLS21]). *Let $(L_t^2)_{\infty < t < 0}$ be an exact, almost calibrated LMCF with finite time singularity with some tangent flow $P_1 \cup P_2$ such that $P_1 \cap P_2 = \{0\}$. Then the following holds*

1. *The tangent flow is unique.*
2. *Lagrangian angle of P_1 is same as the Lagrangian angle of P_2 .*
3. *The type II blow up is Lawlor neck with blowdown $P_1 \cup P_2$.*

Case 2 $P_1 \cap P_2 = \ell$ i.e intersection is a line which we call ℓ .

Theorem 3.18 (Lambert-Lotay-Schulze, [LLS21]). *Let $L_t^2 \hookrightarrow \mathbb{C}^2$ be an exact, almost calibrated LMCF with finite time singularity with blowdown $P_1 \cup P_2$ such that $P_1 \cap P_2 = \ell$, then $L_t = L + tv$ for some constant $v \in \ell$, i.e L is a translator, whcih implies the blowdown is unique.*

Case 3 The two planes P_1 and P_2 are same, i.e we get a single plane P with multiplicity 2.

Theorem 3.19 (Lambert-Lotay-Schulze, [LLS21]). *Suppose that $(L_t^2)_{-\infty < t < 0}$ is an ancient, exact, almost calibrated LMCF with blowdown $2P$, then either*

1. *$L_t = P$ for all $t < 0$ or*
2. *upto rigid motion L_t is a special Lagrangian given as follows*

$$L_t = L = \{(x_1 + i(x_1^2 - x_2^2), x_2 - 2ix_1x_2) \mid x_1, x_2 \in \mathbb{R}\} \quad (20)$$

4 Lecture IV: Conjectures and Open Problems

4.1 Lagrangian and symplectic topology

Lagrangians, in a Riemannian way (curvature), are fundamental objects in symplectic topology. There is a relation between LMCF and symplectic topology.

$$L, L' \hookrightarrow M \text{ Calabi–Yau (e.g. } \mathbb{C}^n)$$

Then

$$L \text{ and } L' \text{ are Hamiltonian isotopic} \iff \exists \text{ a path of Lagrangians } \{L_s\}_{s \in [0,1]}$$

such that

$$L_0 = L, \quad L_1 = L',$$

and

$$\frac{\partial L_s}{\partial s} = J \nabla h_s,$$

where

$$h_s : L_s \longrightarrow \mathbb{R}$$

is a Hamiltonian function.

To denote Hamiltonian isotopy, we write

$$L \sim_{\text{Ham}} L'.$$

If we have *zero Maslov*,

$$\frac{\partial L_t}{\partial t} = H = J \nabla \theta,$$

then the Lagrangian mean curvature flow is a Hamiltonian isotopy.

Conjecture 4.1. *Let $(L_t)_{t \geq 0}$ be a zero-Maslov Lagrangian mean curvature flow in a Calabi–Yau manifold. Since the flow is generated by*

$$H = J \nabla \theta,$$

the flow remains in the Hamiltonian isotopy class of the initial Lagrangian.

Can one exploit this fact to relate the behaviour of the LMCF to invariants from symplectic topology (e.g. the Hamiltonian isotopy class)?

Motivation for the Conjecture

Suppose that I have a Calabi–Yau manifold (M, ω, Ω) . There is a mysterious operation, *Mirror Symmetry* (coming from string theory), which associates another Calabi–Yau manifold

$$(\check{M}, \check{\omega}, \check{\Omega}).$$

Under mirror symmetry, the symplectic geometry of (M, ω, Ω) gets exchanged with the complex geometry of $(\check{M}, \check{\omega}, \check{\Omega})$.

There are many examples.

Symplectic geometry	$\Longleftrightarrow$	Complex geometry
Zero Maslov Lagrangian L	$\Longleftrightarrow$	Holomorphic vector bundle E
Special Lagrangian	$\Longleftrightarrow$	Holomorphic vector bundle E with a Hermitian–Yang–Mills connection

This is the Hitchin–Kobayashi correspondence, proved by Donaldson, Uhlenbeck and Yau.

$$E \text{ admits a Hermitian–Yang–Mills connection} \iff E \text{ is stable.}$$

Comment 4.2. Here “stable” means stable in the sense of algebraic geometry.

Remark 4.3. For irreducible bundles, this theorem is true. The proof uses the Hermitian–Yang–Mills flow.

4.2 Thomas–Yau conjecture

Understanding the Lagrangian Mean Curvature Flow on general manifolds is a big open questions. A particular case, the Calabi–Yau manifolds, exhibits a remarkable property about minimal Lagrangians that are the same as special Lagrangians.

A standing conjecture of Thomas [Tho01], motivated by mirror symmetry, asserts that there is a stability condition for compact graded Lagrangians in a Calabi–Yau manifold, which is expected to determine the existence (and uniqueness) of a special Lagrangian in a given Hamiltonian isotopy class.

Conjecture 4.4. Let $L \hookrightarrow M$ be a compact Calabi–Yau manifold and suppose that L is an almost calibrated Lagrangian.

If L is stable, then the Lagrangian Mean Curvature Flow $\frac{\partial L_t}{\partial t} = H_t$ exists for all time (assuming everything remains smooth), and $L_t \rightarrow L_\infty$, where L_∞ is a special Lagrangian satisfying $L_\infty \sim_{\text{Ham}} L$.

In [LotayOliveira2024DG] Lotay and Oliveira proved the Thomas Conjecture for circle–invariant Lagrangian in a class of hyperkähler 4-manifolds.

In 2002 Thomas and Yau [TY02] stated the following conjecture.

Conjecture 4.5 (Thomas–Yau Conjecture). *If L satisfies either of the conditions:*

- *If the variation in L ’s phase function θ is sufficiently small for all graded sums.*
- *If the Riemannian volume of our Lagrangian L is less than the cohomological volume of any decomposition.*

Then, the mean curvature flow for L exists for all time and converges to a special Lagrangian in its hamiltonian deformation class.

Comment 4.6. • *The statement of the Thomas–Yau conjecture is not completely precise, as the particular stability condition is not yet known.*

- *Almost calibrated is stronger than zero Maslov: the Lagrangian angle takes values in a single interval.*
- $\mathbb{C}^n$ *has a drawback no almost calibrated lagrangian exists*

Example 4.7 (Gibbons–Hawking Ansatz [GH78]). *Choose points $p_1, \dots, p_n \in \mathbb{R}^3$.*

Let

$$V : \mathbb{R}^3 \setminus \{p_i\} \longrightarrow (0, \infty)$$

be a positive harmonic function away from the points p_i , $\Delta V = 0$.

For example,

$$V = m + \sum_{i=1}^n \frac{1}{2|x - p_i|}, \quad m \geq 0,$$

where the summands are Green’s functions. Suppose there exists a vector field X on $\mathbb{R}^3 \setminus \{p_i\}$ such that

$$\operatorname{curl}(X) = \nabla V.$$

Then one can construct a Calabi–Yau metric.

1. *Let ξ be the 1-form dual to X . Take the principal S^1 -bundle*

$$e^{i\psi} \in S^1,$$

and define

$$\eta = d\psi + \xi.$$

Here η is the connection 1-form, determining the horizontal and vertical distributions.

Its curvature satisfies

$$d\xi = *dV,$$

which is the monopole equation.

2. *Define the total space*

$$\hat{M} \xrightarrow{\pi} \mathbb{R}^3 \setminus \{p_i\},$$

which is an S^1 -bundle. Equip it with the metric

$$g = V^{-1}\eta^2 + V g_{\mathbb{R}^3}.$$

The metric is Ricci-flat, hence Calabi–Yau.

3. *If I choose V appropriately, then I can extend the metric g to a manifold*

$$\pi : \widehat{M}^4 \longrightarrow \mathbb{R}^3,$$

where

$$\pi^{-1}(p_i) = P_i.$$

Then

$$\omega = \eta \wedge dx_1 + V dx_2 \wedge dx_3,$$

and

$$\Omega = \left(\eta \wedge dx_2 + V dx_3 \wedge dx_1 \right) + i \left(\eta \wedge dx_3 + V dx_1 \wedge dx_2 \right).$$

4. I can add harmonic functions and obtain a new Calabi–Yau metric.

Exercise 4.8. Let $L \subset \widehat{M}$ be S^1 -invariant. Then

$$L = L_\gamma = \pi^{-1}(\gamma),$$

where $\gamma \subset \mathbb{R}^3$ is a curve.

Show that

$$L_\gamma \text{ is Lagrangian} \iff \gamma \text{ is contained in some plane } \{x_i = \text{const}\}.$$

Moreover,

$$L_\gamma \text{ is special Lagrangian} \iff \gamma \text{ is a straight line.}$$

Finally, show that

$$L_\gamma \text{ satisfies the Lagrangian Mean Curvature Flow} \iff \gamma \text{ satisfies } \frac{\partial \gamma}{\partial t} = V^{-1} k \nu,$$

where V is the Gibbons–Hawking function, k is the curvature of γ , and ν is the unit normal vector to γ .

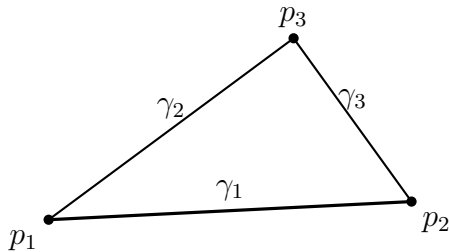
Example 4.9. Suppose

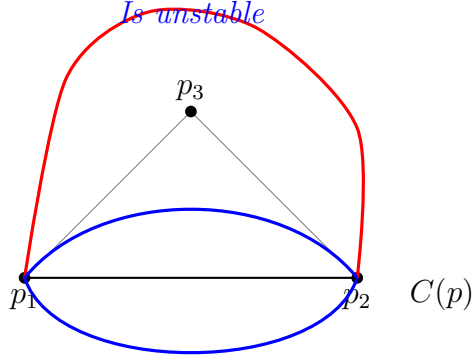
$$p_1, p_2, p_3 \in \mathbb{R}^3$$

lie in the plane

$$\{x_2 = 0\}.$$

Geometrically, the slope of the line determines the Lagrangian angle.





Almost calibrated

Suppose $L \subset M$ is compact, zero Maslov, and almost calibrated. Then

$$L \text{ stable} \iff \forall L_1, L_2 \text{ compact, almost calibrated,}$$

with

$$L \sim_{\text{Ham}} L_1 \# L_2,$$

we have

$$(a) [\arg \Omega|_{L_1}, \arg \Omega|_{L_2}] \not\subset [\inf_L \theta, \sup_L \theta].$$

If it were lines then we have special Lagrangians L_1 and L_2 with $\arg|_{L_i} \Omega = \Theta_i$

$$(b) \text{Vol}(L) < \left| \int_{L_1} \Omega \right| + \left| \int_{L_2} \Omega \right| < \text{Vol}(L_1) + \text{Vol}(L_2).$$

If we have that they are special Lagrangians then we have equality in the above inequality.

Remark 4.10. • To obtain an unstable example, one has to violate both conditions.

• Moreover, the Maslov number has to satisfy a condition involving the winding number.

4.2.1 Joyce Conjecture

There is an upgrade of this statement, we no longer assume that L is almost calibrated; only that it is compact and has zero Maslov class, namely the **Joyce Conjecture**. Joyce [Joyce] updated the Thomas and Thomas–Yau conjectures by making use of the notion of Bridgeland stability condition, now seeking, in good cases, to find a sum of special Lagrangians representing a given Hamiltonian isotopy class. As a way to tackle this conjecture, Joyce proposed the use of Lagrangian mean curvature flow through finite time singularities (possibly with surgeries).

Conjecture 4.11 (Joyce Conjecture [Joy]). *Let $L \subset M$ be a compact, zero-Maslov Lagrangian in a Calabi–Yau manifold. Let $L \subset M$ be a Lagrangian in a Calabi–Yau manifold. Then the Lagrangian Mean Curvature Flow starting at L develops only finitely many singularities, and*

$$L_t \longrightarrow L_\infty = L_1 \cup \cdots \cup L_m,$$

where each L_j is a special Lagrangian. Moreover, this decomposition is canonical.

Then, the version stated as Principle 3.1(a), Problem 3.12(a) and Conjecture 3.16(ii)-(iii) in [Joy] has been proved by Lotay and Oliveira [LO24a].

Theorem 4.12 (Lotay–Oliveira [LO24a], Theorem 1.1 and 1.2). *The Thomas–Yau conjecture holds for S^1 -invariant Lagrangian mean curvature flow in Gibbons–Hawking manifolds.*

This is expected to be equivalent to the Harder–Narasimhan Filtration, under suitable topological assumptions.

References

- [Anc] H. Anciaux. “Construction of Lagrangian self-similar solutions to the mean curvature flow in $\mathbb{C}^n$ ”. In: *Geom. Dedicata* 1 ().
- [CM12] Tobias H Colding and William P Minicozzi. “Generic mean curvature flow I; generic singularities”. In: *Annals of mathematics* (2012), pp. 755–833.
- [CU99] I. Castro and F. Urbano. “On a minimal Lagrangian submanifold of $\mathbb{C}^n$ foliated by spheres”. In: *Michigan Math. J.* 46.1 (1999).
- [ELS20] C.G. Evans, J.D. Lotay, and F. Schulze. “Remarks on the self-shrinking Clifford torus”. In: *Journal für die Reine und Angewandte Mathematik* 758.1 (2020).
- [GH78] G.W. Gibbons and S.W. Hawking. “Gravitational multi-instantons”. In: *Physics Letters B* 78.4 (1978), pp. 430–432. ISSN: 0370-2693. DOI: [https://doi.org/10.1016/0370-2693\(78\)90478-1](https://doi.org/10.1016/0370-2693(78)90478-1). URL: <https://www.sciencedirect.com/science/article/pii/0370269378904781>.
- [Gro07] K. Groh. “Singular behaviour of equivariant Lagrangian mean curvature flow”. PhD thesis. Gottfried Wilhelm Leibniz Universität Hannover, 2007.
- [Gro85] Mikhail Gromov. “Pseudo holomorphic curves in symplectic manifolds”. In: *Inventiones Mathematicae* 82.2 (1985), pp. 307–347.
- [GSS+07] K. Groh, M. Schwarz, K. Smoczyk, and K. Zehmisch. “Mean Curvature Flow of Monotone Lagrangian Submanifolds”. In: *Mathematische Zeitschrift* 257.2 (2007), pp. 295–327. ISSN: 1432-1823. DOI: [10.1007/s00209-007-0126-3](https://doi.org/10.1007/s00209-007-0126-3). URL: <https://doi.org/10.1007/s00209-007-0126-3>.
- [HS99] Gerhard Huisken and Carlo Sinestrari. “Mean curvature flow singularities for mean convex surfaces”. In: *Calculus of Variations and Partial Differential Equations* 8.1 (1999), pp. 1–14.
- [Hui90] Gerhard Huisken. “Asymptotic behavior for singularities of the mean curvature flow”. In: *Journal of Differential Geometry* 31.1 (1990), pp. 285–299.
- [Joy] Dominic Joyce. “Conjectures on Bridgeland stability for Fukaya categories of Calabi–Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow”. In: *EMS Surv. Math. Sci.* 2.1 ().
- [LLS21] Ben Lambert, Jason D. Lotay, and Felix Schulze. “Ancient solutions of Lagrangian mean curvature flow”. In: *ANNALI SCUOLA NORMALE SUPERIORE - CLASSE DI SCIENZE* 22.3 (2021), pp. 1169–1205.

- [LO24a] Jason D. Lotay and Gil Oliveira. “Neck pinch singularities and Joyce conjectures in Lagrangian mean curvature flow with circle symmetry”. In: *Journal of the European Mathematical Society* 24.2 (2024).
- [LO24b] Jason D. Lotay and Gonalo Oliveira. “Special Lagrangians, Lagrangian mean curvature flow and the Gibbons-Hawking ansatz”. In: *Journal of Differential Geometry* 126.3 (2024), pp. 1121–1184. DOI: [10.4310/JDG/1717348872](https://doi.org/10.4310/JDG/1717348872). arXiv: [2002.10391](https://arxiv.org/abs/2002.10391) [math.DG].
- [Nev] A. Neves. “Finite time singularities for Lagrangian mean curvature flow”. In: *Annals of Mathematics* 177.2 ().
- [Nev07] Andr  Neves. “Singularities of Lagrangian mean curvature flow: zero-Maslov class case”. In: *Inventiones mathematicae* 168.3 (2007), pp. 449–484.
- [Nev10] Andr  Neves. “Singularities of Lagrangian Mean Curvature Flow: Monotone case”. In: *Mathematical Research Letters* 17.1 (2010).
- [Nev13] Andr  Neves. “Finite time singularities for Lagrangian mean curvature flow”. In: *Annals of Mathematics* 177.2 (2013).
- [Smo96] Knut Smoczyk. “A canonical way to deform a Lagrangian submanifold”. In: *arXiv preprint dg-ga/9605005* (1996).
- [SS19] Andreas Savas-Halilaj and Knut Smoczyk. “Lagrangian mean curvature flow of Whitney spheres”. In: *Geometry & Topology* 23.2 (2019), pp. 1057–1084.
- [Su19] W.-B. Su. “Mean curvature flow of asymptotically conical Lagrangian submanifolds”. In: *Transactions of the American Mathematical Society* 373.2 (2019).
- [Tho01] R. P. Thomas. “Moment maps, monodromy and mirror manifolds”. In: *Symplectic geometry and mirror symmetry* 467–498 (2001).
- [TY02] R. P. Thomas and S.-T. Yau. “Special Lagrangians, stable bundles and mean curvature flow”. In: *Communications in Analysis and Geometry* 10.5 (2002), pp. 1075–1113.
- [Via21] C. Viana. “A note on the evolution of the Whitney sphere along mean curvature flow”. In: *Journal of Geometric Analysis* 4240–4252.31 (2021).
- [Woo24] Albert Wood. “Singularities of Equivariant Lagrangian Mean Curvature Flow”. In: *Communications in Analysis and Geometry* 32.7 (2024).